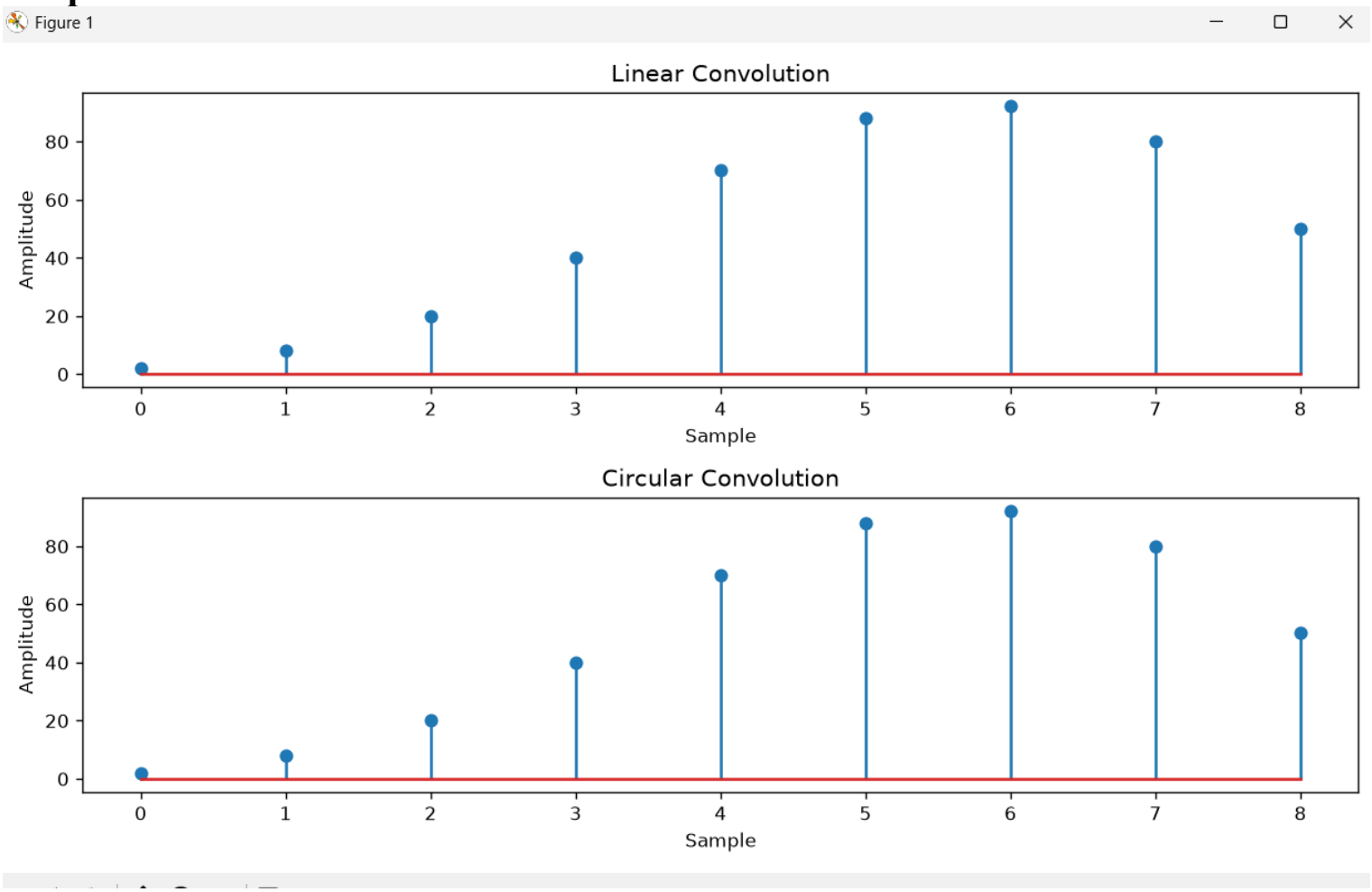
Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology**	
Subject: DSIP	**Aim:** Simulate linear convolution and circular convolution on discrete time signals.	
Experiment:-2	**Date:22-07-2026**	**Enrollment No:92400133057**

Code:

```
import numpy as np
import matplotlib.pyplot as plt
def linear_convolution(signal1, signal2):
    linear_conv = np.convolve(signal1, signal2, mode='full')
    return linear_conv
def circular_convolution(signal1, signal2):
    # Compute the circular convolution
    fft_length = len(signal1) + len(signal2) - 1
    fft_signal1 = np.fft.fft(signal1, fft_length)
    fft_signal2 = np.fft.fft(signal2, fft_length)
    circular_conv = np.fft.ifft(fft_signal1 * fft_signal2)
    return circular_conv
# Define the discrete-time signals
signal1 = np.array([1, 2, 3, 4, 5])
signal2 = np.array([2, 4, 6, 8, 10])
# Compute the linear convolution
linear_conv = linear_convolution(signal1, signal2)
# Compute the circular convolution
circular_conv = circular_convolution(signal1, signal2)
# Plot the linear and circular convolution results
plt.figure(figsize=(10, 6))
plt.subplot(2, 1, 1)
plt.stem(linear_conv)
plt.title('Linear Convolution')
plt.xlabel('Sample')
plt.ylabel('Amplitude')
plt.subplot(2, 1, 2)
plt.stem(circular_conv)
plt.title('Circular Convolution')
plt.xlabel('Sample')
plt.ylabel('Amplitude')
plt.tight_layout()
plt.show()
```

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Output:



Conclusion:

This experiment shows that circular and linear convolution produce the exact same results if you add enough zeros to the signals.

Because we set the FFT length to 9 instead of 5, we gave the signals enough room so they wouldn't wrap around and overlap with themselves. That is why both graphs in the output look completely identical.